

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

for

Social Networking for College Students

A College-Focused Social Networking Platform

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Project Title	Social Networking for College Students
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Primary Users	College Students
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1. Introduction

1.1 Purpose

This Software Requirements Specification defines the functional and non-functional requirements for Social Networking for College Students, a web-based platform designed specifically for students to connect, communicate, collaborate and share information within their college community.

1.2 Document Conventions

Functional requirements use FR-XX and non-functional requirements use NFR-XX. The keyword “shall” indicates a mandatory requirement. Priority levels are High, Medium and Low.

1.3 Intended Audience

This document is intended for the student/development team, project supervisor, faculty evaluator, UI/UX designer and software tester.

1.4 Project Scope

The system provides student registration and login, profiles, connections, posts, comments, likes, private messaging, groups, events, notifications, academic resource sharing and search for students, groups and posts.

1.5 References

- Project description and requirements supplied for Social Networking for College Students.
- IEEE Std 830-1998 – Recommended Practice for Software Requirements Specifications.
- ISO/IEC/IEEE 29148:2018 – Systems and Software Engineering – Requirements Engineering.

2. Overall Description

2.1 Product Perspective

The system is a college-focused social networking platform that brings student communication and collaboration activities into one place. It can operate as a standalone web application and may later integrate with college services.

2.2 Product Features

- Student registration and secure login.
- Student profile creation and editing.
- Add/connect with other students.
- Create, like, comment on and share posts.
- Private messaging.
- College groups and communities.
- Event creation and participation.
- Notifications.
- Academic resource sharing.
- Search for students, groups and posts.

2.3 User Classes and Characteristics

User Class	Description	Main Activities
Student	Primary platform user.	Profile, connections, posts, messages, groups, events and res
Organizer	Student managing a group or event.	Create/manage group or event information.
Administrator (future)	Platform management role.	Manage users, content and moderation.

2.4 Operating Environment

The application is intended for modern browsers such as Chrome, Firefox, Edge and Safari on desktops, laptops, tablets and smartphones.

2.5 Constraints

- Secure handling of student account information is required.
- Messaging, notifications and resource sharing depend on backend implementation.
- Storage and moderation requirements increase with user growth.

2.6 Assumptions and Dependencies

- Users have valid student information.
- Users have internet connectivity.
- A backend and database will be available for a complete implementation.

3. Functional Requirements

3.1 Registration and Login

ID	Requirement	Priority
FR-01	Allow student account registration.	High
FR-02	Allow registered students to log in securely.	High
FR-03	Provide logout functionality.	High

3.2 Student Profile

ID	Requirement	Priority
FR-04	Allow students to create and edit profiles.	High
FR-05	Display relevant information such as name, course and interests.	High

3.3 Connections

ID	Requirement	Priority
FR-06	Allow students to search for other students.	High
FR-07	Allow students to send, accept or reject connection requests.	High
FR-08	Display a student's connections.	Medium

3.4 Posts, Likes, Comments and Sharing

ID	Requirement	Priority
FR-09	Allow students to create posts.	High
FR-10	Allow students to like and comment on posts.	High
FR-11	Allow students to share posts where permitted.	Medium

3.5 Private Messaging

ID	Requirement	Priority
FR-12	Allow connected students to send private messages.	High
FR-13	Display user conversations and unread messages.	High

3.6 Groups and Communities

ID	Requirement	Priority
FR-14	Allow students to discover college groups.	Medium
FR-15	Allow authorized users to create groups.	Medium
FR-16	Allow students to join or leave groups.	Medium

3.7 Events

ID	Requirement	Priority
FR-17	Allow authorized users to create college-related events.	Medium
FR-18	Display event details.	High

ID	Requirement	Priority
FR-19	Allow students to participate/register for events.	Medium

3.8 Notifications

ID	Requirement	Priority
FR-20	Notify users about connection requests, messages and interactions.	High
FR-21	Display notifications in a dedicated area.	Medium

3.9 Academic Resources

ID	Requirement	Priority
FR-22	Allow students to share academic resources.	Medium
FR-23	Allow students to access shared resources.	Medium

3.10 Search

ID	Requirement	Priority
FR-24	Allow searching for students.	High
FR-25	Allow searching for groups and posts.	Medium
FR-26	Present search results in an organized manner.	Medium

4. External Interface Requirements

4.1 User Interface

The platform should provide a clear dashboard, navigation menu, profile pages, feed, messaging interface, groups, events, notifications and search. Buttons, icons and labels should provide clear feedback.

4.2 Hardware Interface

No specialized hardware is required. The system should work on standard computers, laptops, tablets and smartphones.

4.3 Software Interface

The application requires a modern web browser. A complete implementation may use a frontend framework, backend server, database and authentication service.

4.4 Communication Interface

Secure network communication should use HTTPS, especially for login, messages and student information.

5. Use Case Summary

Actor	Use Case	Description
Student	Register/Login	Create an account and access the platform.
Student	Manage Profile	Create and update profile information.
Student	Connect	Find classmates and manage connection requests.
Student	Create/Interact with Posts	Create, like, comment and share posts.
Student	Message	Send and receive private messages.
Student	Join Groups	Discover and participate in communities.
Student	Participate in Events	View and participate in college events.
Student	Share Resources	Share or access academic resources.
Student	Search	Find students, groups and posts.
Administrator	Manage Platform	Future administration and moderation.

6. Non-Functional Requirements

6.1 Performance

- Main pages should load within an acceptable time.
- Common actions should respond without noticeable delay.
- The system should support increasing numbers of students through appropriate backend design.

6.2 Security

- Passwords shall be securely hashed.
- Authentication and student data communication shall use HTTPS.
- Private messages and account information shall be accessible only to authorized users.
- User input should be validated.

6.3 Usability

- The interface should be simple for students to learn.

- Navigation should use understandable labels and icons.
- The interface should be responsive.

6.4 Reliability

The system should preserve user accounts and important platform data reliably and handle common failures gracefully.

6.5 Maintainability

The software should use modular components so features such as groups, messaging, events and notifications can be maintained independently.

6.6 Scalability

The system should support additional students, groups, posts, messages and events without major architectural changes.

7. Software Testing Considerations

Test ID	Requirement	Test Scenario	Expected Result
TC-01	FR-01	Register a new student	Account is created.
TC-02	FR-02	Login with valid credentials	Student is authenticated.
TC-03	FR-04	Create/edit profile	Updated information is displayed.
TC-04	FR-07	Send connection request	Request reaches selected student.
TC-05	FR-09	Create a post	Post appears in feed.
TC-06	FR-10	Like/comment on post	Interaction is recorded.
TC-07	FR-12	Send private message	Message appears in conversation.
TC-08	FR-18	Open event	Event details display correctly.
TC-09	FR-20	Trigger notification	Relevant notification appears.
TC-10	FR-24	Search student	Relevant results are displayed.

Recommended testing includes functional, UI, responsive, authentication/security, database, messaging and browser compatibility testing.

8. Proposed System Architecture

8.1 Architecture Overview

A complete implementation can follow a three-layer architecture. The presentation layer handles the interface. The application layer manages authentication, posts, connections, messaging, groups, events and notifications. The data layer stores users and platform information.

8.2 Suggested Technology Stack

The supplied project description does not specify an exact technology stack. A possible implementation may use HTML/CSS/JavaScript or a frontend framework, a backend web framework, REST APIs and a relational or NoSQL database. The final stack can be selected according to project requirements.

9. Data Model Overview

Entity	Possible Key Attributes
Student/User	userId, name, email, passwordHash, course, profile information
Connection	connectionId, requesterId, receiverId, status
Post	postId, userId, content, createdAt
Comment	commentId, postId, userId, content
Like	likeId, postId, userId
Message	messageId, senderId, receiverId, content, sentAt, readStatus
Group	groupId, name, description, creatorId
GroupMember	groupId, userId, joinedAt, role
Event	eventId, title, description, organizerId, date, location
EventParticipant	eventId, userId, status
Resource	resourceId, userId, title, description, resourceUrl

Entity	Possible Key Attributes
Notification	notificationId, userId, type, message, readStatus

10. Future Enhancements

- College email/ID-based verification.
- Android/iOS mobile application.
- Real-time chat and push notifications.
- Advanced moderation and reporting.
- AI-assisted recommendations for groups, events and resources.
- College club and society management.
- Integration with official college calendars.
- Advanced privacy controls.
- Administrator analytics dashboard.

11. Glossary

Term	Definition
SRS	Software Requirements Specification.
UI	User Interface.
Authentication	Process of verifying a user's identity.
Authorization	Process of determining permitted access.
Post	Content shared by a student.
Connection	Relationship/request between students.
Group	Community based on a college interest or activity.
Event	College-related activity students can discover and join.
Notification	Alert about relevant platform activity.
HTTPS	Secure protocol for client-server communication.

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End of Document – SRS v1.0